

UCT402: DATATBASE MANGEMENT SYSTEMS

L	T	P	Cr
3	0	2	4.0

Course Objectives: Emphasis is on the need of information systems. Main focus is on E-R diagrams, relational database, concepts of normalization and de-normalization and SQL commands.

Introduction: Introduction to Database, Hierarchical, Network and Relational Models.

Database system architecture: Data Abstraction, Data Independence, Data Definition Language (DDL), Data Manipulation Language (DML).

Data models: Entity-relationship model, network model, relational and object oriented data models, integrity constraints, data manipulation operations.

Relational query languages: Relational algebra, Tuple and domain relational calculus, SQL3, DDL and DML constructs, Open source and Commercial DBMS - MYSQL, ORACLE, DB2, SQL server.

Relational database design: Domain and data dependency, Armstrong's axioms, Functional Dependencies, Normal forms, Dependency preservation, Lossless design.

Query processing and optimization: Evaluation of relational algebra expressions, Query equivalence, Join strategies, Query optimization algorithms.

Storage strategies: Indices, B-trees, Hashing.

Transaction processing: Concurrency control, ACID property, Serializability of scheduling, Locking and timestamp based schedulers, Multi-version and optimistic Concurrency Control schemes, Database recovery.

Database Security: Authentication, Authorization and access control, DAC, MAC and RBAC models, Intrusion detection, SQL injection.

Advanced topics: Object oriented and object relational databases, Logical databases, Web databases, Distributed databases, Data warehousing and data mining.

Laboratory work:

Students will perform SQL commands to demonstrate the usage of DDL and DML, joining of tables, grouping of data etc. They will also implement one project.

Project:

It will contain database designing & implementation, should be given to group of 2-4 students. While doing projects emphasis should be more on back-end programming like use of SQL. Project should have continuous evaluation and should be spread over different components.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

On completion of this course, the students will be able to:

1. Analyze the basics of Database management Systems, its need and advantages as compared to traditional file-based systems.
2. Analyze and design of database using data models like E-R by identifying entities, attributes and relationships.
3. Demonstrate use of the SQL with Data Definition Language (DDL), Data Manipulation Language (DML).
4. Apply and create Relational Database Design process with Normalization and Denormalization of data.
5. Comprehend the concepts of transaction management, concurrency control and recovery management.

Text Books:

1. Database System Concepts. Abraham Silberschatz, Henry F. Korth and S. Sudarshan.

Reference Books:

1. Principles of Database and Knowledge Base Systems, Vol 1 by J. D. Ullman.
2. Fundamentals of Database Systems. R. Elmasri and S. Navathe.
3. Foundations of Databases. Serge Abiteboul, Richard Hull, Victor Vianu.